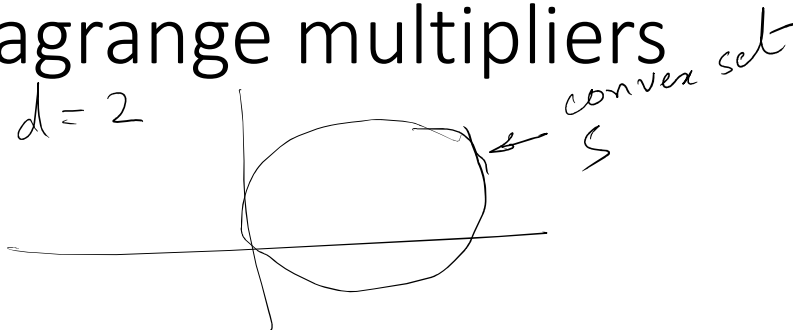
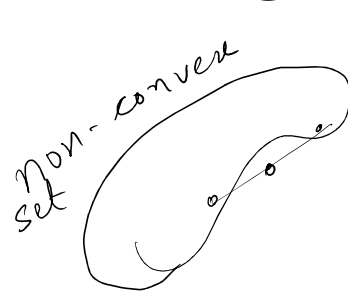


Constrained optimization using Lagrange multipliers

$$\begin{aligned} & \min_W F(W) \\ & W \in \mathbb{R}^d \text{ such that } W \in S \end{aligned}$$



$S \equiv$ convex set if + when w^1 & $w^2 \in S$ then
 $\lambda w^1 + (1-\lambda)w^2 \in S \quad \forall \quad 0 \leq \lambda \leq 1.$

Examples of convex set

$$1) \quad \|w\|_2^2 \leq \gamma \quad \equiv \quad \sum_{j=1}^d w_j^2 \leq \gamma$$

$$2) \quad \|w\|_1 \leq \gamma \quad \text{is also convex set}$$

$$3) \quad w^T c \leq a \quad \text{where } c \in \mathbb{R}^d$$

$$4) \quad w^T c = 1$$

More examples of convex set

If $g(w)$ is convex then $g(w) \leq 0$ is ~~a~~ convex set. S

Proof:

$g(w)$ is convex. If w^1, w^2 are such that $g(w^1) \leq 0$ & $g(w^2) \leq 0$
i.e., $w^1, w^2 \in S$ then
 $\lambda w^1 + (1-\lambda)w^2$ also belongs to S $\forall 0 \leq \lambda \leq 1$
since
 $g(\lambda w^1 + (1-\lambda)w^2) \leq \lambda g(w^1) + (1-\lambda)g(w^2)$ [property of convex functions]
 ≤ 0 since $w^1, w^2 \in S$.

But $g(w)$ does not have to be convex for $g(w) \leq 0$ to define a convex w set.

e.g. $g(w) = -e$ $\nrightarrow$ not convex.

But $g(w) \leq 0 \in \mathbb{R} \nrightarrow$ convex.

We will consider the case of such constrained optimization problem

$$\min_w F(w)$$

s.t.

$$g_i(w) \leq 0 \quad i = 1 \text{ to } k \quad \text{and} \quad g_i(w) \text{ is convex in } w.$$

Consider the case where $k=1$

$$\begin{array}{l} \min_w F(w) \\ \text{s.t. } g(w) \leq 0 \end{array}$$

Another way to express the above:

$$\rightarrow \min_w F(w) + \infty \mathbb{I}(g(w) > 0)$$

Replace this objective with a LIB:

$$\begin{array}{l} \min_w F(w) + u g(w) \\ u \geq 0 \end{array}$$

where $\mathbb{I}(\sigma)$ is an indicator function that is 1 if σ is true and 0 otherwise

$$\min_w \left(\max_{u \geq 0} F(w) + u g(w) \right) = \min_w L(w, u)$$

$$\Leftrightarrow \min_w F(w) \quad \text{s.t. } g(w) \leq 0$$

$$L(w) = \max_{u \geq 0} L(w, u) =$$

$$\begin{cases} \infty & \text{if } g(w) > 0, u^* \rightarrow \infty \\ F(w) & \text{if } g(w) \leq 0, u^* = 0 \end{cases}$$

$$\Rightarrow \min_w \max_{u \geq 0} L(w, u) \Leftrightarrow$$

$$\min_w F(w) \quad \text{s.t. } g(w) \leq 0$$

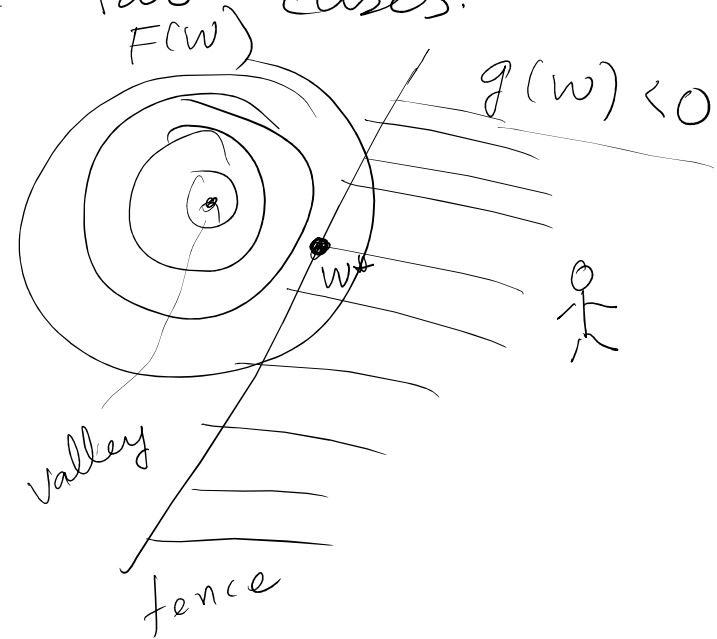
Conditions for optimality of constrained optimization.

At $(\underline{w}^*, \underline{u}^*)$ that is optimal we have two cases.

- 1) $\min_w F(w)$ does that satisfy the constraint.

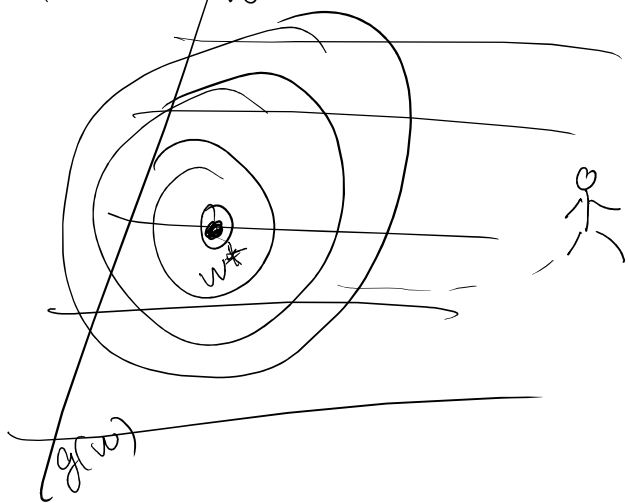
The optimum will be at the boundary.

At w^* $g(w^*) = 0$



- 2) $\min_w F(w)$ satisfies the constraints then $u^* = 0$

since $g(w^*) < 0$, and we want to solve for $\max_u L(w, u)$



① Complementary slackness condition
 $u^* \cdot g(w^*) = 0$

when constraint is active:

At w^* there must not exist any direction $\vec{e}$

$$\text{s.t. } \left. \begin{array}{l} \nabla g(w^*) \cdot \vec{e} < 0 \\ \& \nabla F(w^*) \cdot \vec{e} < 0 \end{array} \right\}$$

Farkas's Lemma:

$$\nexists x \text{ s.t. } Ax < 0 \quad \text{then} \quad \exists u \text{ s.t. } A^T u = 0$$

$$\exists e \text{ s.t. } \begin{bmatrix} \nabla g(w^*)^T \\ \nabla F(w^*)^T \end{bmatrix} \begin{bmatrix} e_1 \\ \vdots \\ e_d \end{bmatrix} < 0 \Rightarrow u \geq 0$$

$$\exists u_1 \neq 0 \quad \text{s.t.} \quad u_1 \nabla F(w^*) + u_2 \nabla g(w^*) = 0$$

$$\exists u \geq 0 \Leftrightarrow \nabla F(w^*) + u^* \nabla g(w^*) = 0$$

②

③

$$u^* \geq 0$$

④

$$g(w^*) < 0$$

Combining all conditions we get the KKT conditions:
 and generalizing to $k > 1$
 (u^*, w^*) is a local optima of $\min_w \max_{u \geq 0} F(w) + \sum_{i=1}^k u_i g_i(w)$
 if and only if

- 1) $\nabla_w F(w^*) + \sum_{i=1}^k u_i^* \nabla g_i(w^*) = 0$ ($g(w)$ is differentiable)
- 2) Feasibility conditions: $g_i(w^*) \leq 0$; $u_i^* \geq 0 \forall i=1 \dots k$
- 3) Complementary slackness condition: $u_i^* g_i(w^*) = 0 \forall i=1 \dots k$

If $F(w)$ & $g(w)$ are convex then w^* is a global optima subject to the constraint.

Proximal gradient descent algorithm! (PGD)

- $\min_{\theta} \underbrace{L(\theta) + R(\theta)}_{F(\theta)}$ $\leftarrow$ convex function but not necessarily differentiable.

$L(\theta)$ loss function, differentiable
convex for linear regression

[Sub-gradient descent may be applied but it may not be efficient.]

$\rightarrow$ PGD comprises of two-steps corresponding to one step of GD

1) Take a step in the direction of the gradient of loss $L(\theta)$

$$\tilde{\theta}_{t+1} = \theta_t - \nabla_{\theta} L(\theta_t)$$

2) Project the above "draft" $\tilde{\theta}_{t+1}$ into a space that respect $R(\theta)$

$$\theta^{t+1} = \theta_t - \eta \nabla_{\theta} F(\theta_t)$$

$$\theta_{t+1} \leftarrow \text{Prox}_{\eta_t R(\theta)}[\tilde{\theta}_{t+1}]$$

$$\text{prox}_{\eta R(\theta)}(\tilde{\theta}) = \underset{\theta}{\operatorname{argmin}} \left(\underline{R(\theta)} + \frac{1}{2\eta} \|\tilde{\theta} - \theta\|_2^2 \right)$$

$R(\theta)$ is often simple enough that the above objective can be solved in closed form.

The above framework can be used to solve some "easy" constrained optimization problems.

Projected gradient descent algorithm;

One common type of constraint is box constraints on parameter:

$$l_j \leq \theta_j \leq u_j \quad l_j, u_j \in \mathbb{R};$$

$$\begin{array}{l} \min_{\theta} L(\theta) \\ \text{s.t.} \\ j=1 \dots d: \end{array} \quad l_j \leq \theta_j \leq u_j \quad \Bigg| \quad \Leftrightarrow$$

$$\begin{array}{l} \min_{\theta} L(\theta) \\ \text{s.t.} \\ l_j - \theta_j \leq 0 \\ \theta_j - u_j \leq 0 \end{array} \quad \Bigg\} \quad \forall j$$